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Research Paper

Defining 'adequate' tree protection: Meeting urban canopy targets requires careful retention of mature trees

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HIGHLIGHTS

- Large street trees are shown to be crucial to urban canopy expansion.
- Models show mature tree loss generally should not exceed 1–3% p.a.
- Low removal rates strongly predict canopy growth performance.
- Mature tree loss above 1% makes canopy targets hard to achieve.
- Tree removal modelling supports governance for improved urban forest protection.

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ABSTRACT

Canopy cover is increasingly recognised as an important aspect of the urban landscape, and as a result urban forestry plans have been adopted in many cities globally. These are often built around ambitious canopy cover targets, with a goal to keep cities cool and improve their amenity while delivering co-benefits like health improvements, flood reduction and enhanced biodiversity. Canopy targets and tree planting have tended to receive more attention than another critical determinant of canopy expansion: keeping the trees alive once they are planted. In this study, we use a model to examine how varying tree removal rates influences the long-term canopy cover provided by street trees, considering both typical policy timeframes and projections through the end of the century. The model uses allometrics from real urban trees in four cities to project canopy growth under a wide range of removal rates, for a wide range of potential urban forest characteristics. We find that retention of mature and maturing street trees is consistently critical, with removal rates exceeding 3% per annum generally associated with much lower canopy progress (and usually very poor results). The model's outputs demonstrate how cities can build a clear case for better-integrated urban governance to ensure these challenging threshold values are avoided.

1. Introduction

Governing an urban landscape successfully requires a constellation of actors to deliver a set of challenging built form outcomes in an integrated, well-balanced way. A well-governed urban landscape might, for example, offer efficient transport, reliable utilities, high aesthetic quality and a supportive context for sociability, commerce and civic expression. Balancing these priorities both in design and operations is a central challenge in managing a city's public spaces (Gehl, 2001). Another important role of the urban landscape is providing adequate

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green infrastructure in the places people live and work, both for recreation and to manage climate risks and keep residents healthy (Konijnendijk, 2022; Lawrence et al., 2013; Williams et al., 2025). Recent studies have highlighted how significant this provision needs to be; for example Ziter et al. (2019) found that urban canopy cover exceeding 40 % was necessary to deliver substantial temperature reductions in urban areas during hot weather. Beyond just ensuring that cities provide adequate shade on hot days, metrics now also serve to quantify broader green infrastructure provision for human needs. Konijnendijk (2022) specifies a heuristic guideline of '3 + 30 + 300′ as a bare minimum for urban nature provision for human health, requiring that each home and workplace has a view to 3 trees, 30 % neighbourhood canopy, and a walk of 300 m or less to a local park. To date, these levels of green infrastructure provision are rarely met; a number of studies have found that major cities are falling considerably short of even this minimum standard (Battisti et al., 2024; Croeser et al., 2024; Nieuwenhuijsen et al., 2022). Retrofitting the urban landscape to provide extensive tree cover as well as a sufficient network of parks and natural areas for a city involves a very substantial allocations of land and capital, and requires a correspondingly serious governance commitment to ensuring the upkeep of these assets while also addressing other priorities (e.g. transport, construction, utility provision).

In this study, we focus on one aspect of this challenge: managing tree removal rates to ensure the urban landscape achieves adequate canopy cover. This focus was selected both because canopy shortfalls are a key problem highlighted in the literature, and because tree removal rates are often high in cities. This is posed as an issue of urban landscape governance, rather than a narrow problem for urban forestry teams, because removal rates reflect a very diverse set of drivers and stakeholders which in turn require a well-coordinated governance response.

Removal requests associated with property development are common, both for construction access and when trees conflict with the alignments of new underground infrastructure (Hilbert et al., 2019a; Willis et al., 2024). Accidental tree death during construction or public works maintenance is also a concern, both from heavy machinery and vehicles compressing root zones, and when limbs or major structural roots are severed by careless (or disinterested) contractors (Kirkpatrick et al., 2013). New and upgraded urban infrastructure, including roads, public buildings, recreational spaces, and essential services can necessitate (or minimise) the removal of trees to make way for development. Outside of construction contexts, tree removal requests can arise from residents who dislike ecosystem disservices such as fallen leaves or bird noise, or perceive root intrusions or limb drop from large trees as a threat to their homes or vehicles (Conway, 2016). Tree vandalism also can be problematic in some areas (Esperon-Rodriguez et al., 2023), for example in high pedestrian traffic areas, near alcohol venues and, quite differently, in front of expensive homes with ocean views. Councils themselves can be significant drivers of high removal rates; this can be the result of a fear of liabilities associated with their trees, which in turn may drive heavy use of tree pruning/ pollarding as well as a tendency to remove healthy trees when they display any sign of risk, however manageable that risk might be (Kirkpatrick et al., 2013). Finally, growing conditions in urban settings can be drivers of higher removal rates, as street trees are often planted in constrained soil volumes with very low water and nutrient content, as well as limited rain infiltration and high rates of compaction, surrounded by hot materials that generate challenging microclimates. These factors make establishment of young trees more difficult, and can reduce growth rates and lifespans of those trees that do successfully establish, both due to direct physiological stressors like water shortage and increased susceptibility to pests and diseases (Smith et al., 2019a). Collectively, this constitutes a formidable set of management challenges, which, if not addressed adequately, can quite easily result in tree removal rates that completely negate any progress on canopy cover arising from new plantings.

Collectively, this constitutes a formidable and diverse set of governance challenges, which, if not addressed adequately, can quite easily result in tree removal rates that completely negate any progress on canopy increase arising from new plantings. Accordingly, cities with bold canopy targets and ambitious tree planting programs must have sophisticated management capabilities and stakeholder relationships to ensure the establishment, protection and lasting health of their street trees, both through direct management action and by fostering citizen stewardship (Boyce, 2010).

As cities shift focus from planning to implementation, they face a new management frontier: reducing rates of tree removal. To date, regular removal of trees and replacement with saplings remains common (and poorly monitored) in many cities, even when ambitious canopy targets exist (Croeser et al., 2020; Pretzsch et al., 2021). The relatively limited attention to this crucial operational success factor is striking when compared to the focus on 'up front' questions that tend to occupy authors of urban forest strategies, such as canopy targets and planting rates. Perhaps due to the relatively operational nature of street tree protection (which makes it politically marginal and less rewarding to senior leaders), management of urban forests varies widely in terms of both the effort put into establishing young trees, and the care given to protecting maturing and mature trees. Tree protection remains a relatively data-poor and understudied aspect of urban landscape management (Ko et al., 2015b; Pretzsch et al., 2021). While canopy targets abound, we are not aware of any city that has corresponding targets for tree protection.

Accordingly, our study seeks to define what 'adequate' management of tree removal might be, in quantitative terms. Specifically, we aim to demonstrate how cities can determine how much street tree removal can take place annually before canopy targets are compromised. We examine the impacts of removal rates for young trees, maturing trees and mature trees, both over a typical policy timeframe (2025 to 2055) and to the end of the century (to offer a sense of longer-term management implications). Street trees are selected as a focus because streets typically form a large majority of public space by land area in cities; the practical realisation of tree canopy cover targets via public land actions is by therefore heavily reliant on street tree canopy (Furchtlehner et al., 2022; UN Habitat, 2013).

To derive these insights, we employ scenario modelling to establish the approximate minimum survival rates at which street tree populations can (a) avoid net losses of canopy (b) achieve canopy increase at typically desirable rates. Our model exposes a hypothetical tree population to a series of increasingly high tree removal rates, and tracks canopy increases between 2025 and 2100.

Recognising that urban forests occur in many different urban settings around the world, we also use the scenario model to explore how an array of different contexts respond to varied rates of removal to produce different canopy cover outcomes. This reflects the fact that there is a lot of variety in the biophysical characteristics of street trees, the urban landscapes they exist in, and the ways that they are managed. Some urban forests are mostly mature trees; others, newly-planted. Similar variety exists for species distribution, both in terms of how evenly the species palette is used, and the openness cities have to exotic species use. Management of tree removal, and what happens when a tree is removed, is also highly variable; one city may replant a removed tree immediately, while another may wait two years and then replant the tree as well as one or more additional trees to compensate the net loss of cover. Accordingly, we apply the model to a wide array of management approaches to demonstrate how these influence a city's threshold at which removal rates are likely to become problematic.

While our model is a helpful tool in understanding how best to maintain canopy in the urban landscape, the concept of managing tree populations to sustainably provide a certain ecosystem service at a set level extends back to at least the 18th century, as early foresters recognised the need to optimise tree age distributions within woodlots to ensure reliable yields (e.g. (Von Carlowitz, 1713)). More recent studies have transferred this logic to the modern context of urban forest management, wherein tree populations must be managed to provide

sustainable (and often increasing) ecosystem services such as cooling and stormwater interception. Approaches to projecting ecosystem services at the individual tree scale are demonstrated by Rötzer et al. (2019), using site-specific validated equations to project quite precise benefits for four species in central Europe. At much larger and more general scale, Pretzsch et al. (2021) demonstrated how age distributions in a notional population of thousands of trees can be optimised to ensure sustainable supply of ecosystem services and biophysical traits such as canopy cover, cooling and tree biomass. The latter study notes the potential for future models to include consideration of species-specific allometry (growth equations), local climates, and more varied mortality rates. The model presented in this paper addresses each of these frontiers, employing empirically-derived allometrics (i.e. tree growth equations) that are specific to dozens of individual tree species in four very different US climatic contexts (McPherson et al., 2016). The model's key input variable is the removal rate of trees within the studied population, with 125 different removal rates within the study modelled in total. While this research represents an important progression in each of these areas, its design for the understanding of tree population dynamics to support derivation of removal rate targets differentiates it from the ecosystem service projection focus of prior studies, as well as from established practitioner-facing projection tools like iTree (USDA Forest Service, 2025).

The model is used to answer three key research questions:

1. What removal rates for street trees within each age cohort are associated with achievement of typically desirable canopy increase/retention outcomes?
2. How do the removal rates of different age cohorts impact overall canopy cover?
3. How do 'acceptable' removal rates vary between different urban forests and urban forest management regimes?

From a practical perspective, answers to these questions offer general insights into the maximum levels of removal of young, maturing and mature trees that can take place before cities become likely to miss their canopy targets, while also demonstrating the range of results cities could expect if they applied the model to their own urban forests.

2. Methods

2.1. Model setup and assumptions

To examine the outcomes of various street tree removal rates at different age distributions, and how these outcomes vary in contrasting climate settings, we establish a growth model for a notional population of 1,000 street trees in a generic urban residential setting, representing a local management area such as a suburb or precinct. We model a range of removal rate scenarios that vary in terms of (a) how many trees are removed and (b) the tree age distribution of removals. We are interested in the cumulative impact of different removal rates over time with species and age distributions, and planting strategies, rather than locationally specific scenarios.

In this study we use the term 'removal rates' to refer to all actions and contexts in which street trees are removed – including due to natural end-of-life (i.e. senescence) as well as early removals due to construction, illness, establishment failure and a host of other threats which we discuss in more detail below. We express this rate as the percentage of all trees in a given cohort removed annually, regardless of reasons for removal – overall tree survival is the focus of this paper, rather than examining the impact of a particular threat or management response. We selected the term 'removal rates', rather than 'mortality', to place emphasis on the agency of built environment professions as a key determinant of tree survival in urban settings.

To establish a plausible base case for the model, we adopt a street layout of 18 m, with a road width that falls within the range of US, European and Asian cities, being slightly smaller than the US average and larger than non-US average (Millard-Ball, 2022). This level of specificity is necessary in order to facilitate an estimate of land area (and therefore canopy cover) in a residential streetscape context. The trees are assumed to be planted along a network of streets with a common layout, constituting two traffic lanes and two parking lanes of typical width, plus a footpath of 3.5 m including both a pedestrian walkway and a planted verge. In total the street width is 18 m, distributed as per Fig. 1. We assume a suburb-sized residential area, constituting 100 of these segments, each 100 m long, along which our tree population is distributed.

Planting is assumed to be at an average density of 10 trees/100 m street length, guided by the research of Smart et al.(2020) which identifies this as an achievable planting density in urban areas (this study found that densities ranged between 1.0 and 10.6 trees per 100 m of street length in five cities). Fig. 1 shows the assumed layout of each street segment; trees are spaced 20 m apart, and the design is set to represent a street that has been successfully 'planted up' (i.e. to capacity) in a very conventional way (e.g. trees spaced 20 m apart, 10 m clearances left at intersections for sightlines, no removal of parking or creation of medians).

Thus, with 100 streets, our tree sample is set at 1000 trees, a number selected to minimise extreme variances between model runs due to random age distributions within species cohorts, while remaining computationally manageable. As the study area constitutes 100 streets of 100 m in length at 18 m building-to-building street width, the study area totals 18 ha in street area. We assume that electric cabling is underground (not overhead) and other utility infrastructure (e.g. internet cabling, sewers etc) is appropriately distant from tree planting sites. Accordingly, pruning of tree limbs or roots to maintain infrastructure clearances is not modelled in this study, and the spacing in Fig. 1 is assumed to be consistently achieved.

Readers will note that we refer exclusively to *street* area and *street* canopy. This approach avoids the complications that arise from variations in canopy cover on private land and the different proportions of public versus private land across urban areas. Our study is specifically concerned with canopy cover over streets (rather than total canopy cover across all land types), as our goal is to define appropriate limits for street tree removal rates in order for streets to help neighbourhoods achieve canopy objectives. While promotion and protection of private land canopy is also important, it presents a very different management challenge, where control is more limited and indirect as actions are taken by landholders rather than local governments, and as such is out of scope in this study (Ordóñez-Barona et al., 2021). Similarly, canopy provided by parks is valuable, but falls outside the scope of this study. In some instances, it is plausible that large trees may provide canopy which shades or even partially overlaps private property boundaries; this is not considered a problem as our study seeks to optimise the *contribution* that street trees make to overall neighbourhood canopy cover.

The model commences in 2025 and projects tree growth to 2100. The model's key input is the climate zone and the removal rate scenario, and the key outputs are the age of each tree each year, as well as the trunk thickness and canopy cover for each tree for each year. This enables us to discern trends in total canopy cover over time for different rates and distributions of street tree removal. In total we model a wide range of different removal scenarios, which are described in detail in the following section of our Methods.

While the model runs to the end of the century, urban forest plans tend to be for shorter timeframes, so we focus much of the analysis on the impacts of removal rates on a thirty-year horizon (2025–2055). Accordingly, the overarching goal is to determine the loss rates for each age cohort at which canopy cover (a) 'breaks even', with no loss or increase, (b) grows at a moderate rate (+2.5 %p per decade canopy cover in streets) and (c) grows at a faster pace, as may be required in contexts where rapid catch-up is required (+5%p per decade canopy cover change in streets). These rates, while hypothetical, are set to plausibly

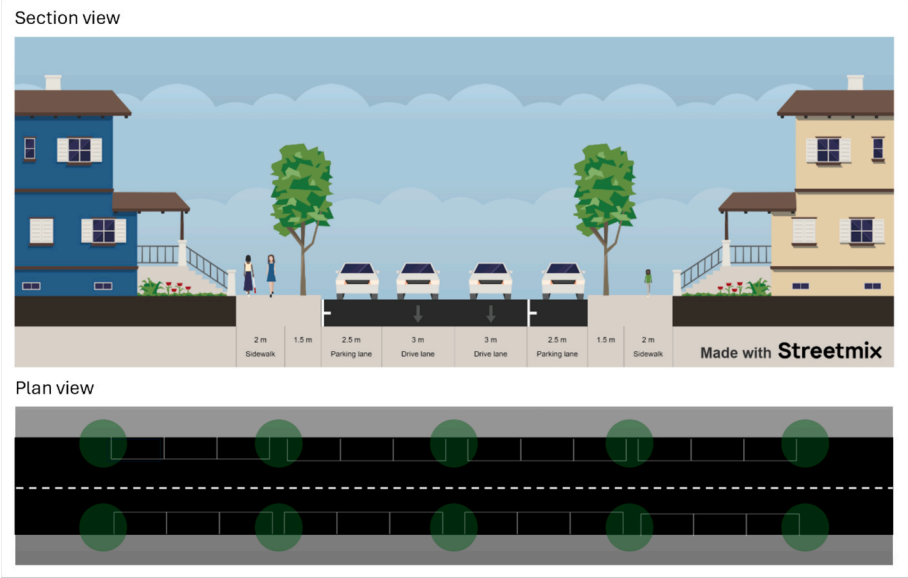


Fig. 1. Street layout assumed in this study. Plan view shows tree planting pattern, which produces a planted density of 10 trees/100 m street length, with 20 m spaces between street trees. Section view prepared using Streetmix – https://streetmix.net/.

approximate three common starting points for urban forest plans, as per Table 1.

Cities with established, healthy urban forests may be satisfied to simply avoid canopy loss or aim for 'slow growth' removal rates, but in many cases more ambitious urban forest targets will require the 'fast' growth rate at a minimum, especially where private green space is declining or rare, or where park distribution is 'clumped' and thus offers few local canopy benefits. Our model serves to identify the removal rates that ensure that each of these growth targets can be met.

As our model's setting is intended to be as generic as possible, the 'location' of the study is not of any special significance. We identified a suitable set of urban tree allometrics (growth formulae) from the United States (McPherson et al., 2016), and chose sets of formulae derived from the diverse cities of Orlando (1367 mm annual rainfall), New York (1041 mm), Berkeley (564 mm) and Albuquerque (250 mm) to use as the species and formulae sets in our model runs. This means that the trees in our model are chosen from a set of species that are common in each of these cities, and have growth characteristics that reflect the climate and built form of that city; this necessary selection of four place-based sets of allometrics is the limit of our study's engagement with particular places.

2.2. Core Model Inputs

In this section we describe each of the key inputs and parameters for the model:

Table 1
Examples of contexts in which our three different management goals might be relevant. Note these are not model inputs, simply contexts in which a city might select one of the three management goals we use as benchmarks. All percentages are absolute (not relative) and represent changes of canopy relative to street area.

Management goal	Context	Starting canopy	Annual gain required to reach 30 % street cover in 30 years
'Avoid loss'	Well-established urban forests	30 %	None (stable canopy)
'Slow growth'	Near-adequate urban forest	22.5 %	0.25 %p
'Fast growth'	Major improvements in canopy required	15 %	0.5 %p

1. The generic street tree population's size, setting and species distribution
2. The allometric equations that were used to model the growth of these trees
3. The range of loss rates modelled for young, maturing and mature trees

2.3. Input 1: tree population size, setting and species distribution

The tree species used in this run of our model are sourced from a large US Forest Service dataset (McPherson et al., 2016), which monitored the growth of 14,000 urban trees over 14 years in a range of US cities. Each city in this dataset includes a set of approximately 20 common local species for which allometric equations exist for urban populations. Our supplementary materials include the list of species used to populate the hypothetical suburb in this study. Our model randomly assigns ages within each cohort at the start of each run; young trees are assigned a time from planting of between 1–5 years, maturing trees are assigned a time from planting between 6–20 years and mature trees are assigned a time-from-planting of 21–60, except where a tree species' maximum age for use in the allometric equations was below 60 years, in which case this maximum was set as the top age for the mature cohort for that species. As street trees are generally pre-grown before planting (Pretzsch et al., 2021), all trees were assumed to be planted after two years of growth in a nursery – so a tree with a 'time from planting' of 2 years is 4 years old in total, for the purpose of allometric calculations. Tree species distribution within each cohort is varied across a range of scenarios, which we ran for each region we modelled.

2.4. Input 2: Allometric equations for tree growth (trunk diameter and canopy change over time)

The USFS dataset discussed above includes allometric equations that quantify how trees grow in terms of diameter at breast height (DBH) and canopy area. Thus, from the age of a tree we are able to quantify canopy cover as a percentage of street area. This canopy cover is dynamic as each tree in our generic population grows, matures, dies due to senescence or artificial removal, and is ultimately replaced one or more times by a sapling.

The urban nature of our allometric dataset is crucial, given allometric equations tend to be for trees in natural or commercial forests, which

can be inaccurate in highly modified urban contexts, where issues like soil compaction, impermeable surface cover and unnatural microclimates make tree growth challenging(Smith et al., 2019b). In this respect the growth rates applied in this study represent a pragmatic picture of the likely outcomes under current practice, rather than the accelerated growth rates that are possible with proper investment in higher-quality planting conditions that allow for greater soil volume and passive watering (Grey et al., 2018; Rötzer et al., 2019).

Notably, the Age-DBH relationship for each species in each region specifies a maximum age beyond which observations of that particular species of tree are unavailable i.e. there were no trees of this age to establish the relationship between age and DBH. The dataset that supplies these equations clearly specifies that this is the maximum age at which the equations should be applied; this is not only because there were not enough observations to be sure of the relationship beyond this point, but also because the formulae can behave unexpectedly if applied outside the recommended age range (e.g. producing impossibly high exponential results).

An important (and conservative) assumption in most of the model runs in this study is that the maximum age in these allometrics represents an age of likely senescence in these settings. This means that trees that reach this age are considered to be overmature, and are removed and replaced by the model automatically, and counted towards the total removal count for mature trees. Recognising this is important point of uncertainty and that some cities may have the capacity to retain trees beyond the ages observed by the allometric studies we use, we model a few variants to the base case where maximum age does not trigger immediate removal (this is discussed in detail later in Methods).

2.5. Input 3: Removal rates for trees, by size, grouped in scenarios for modelling

Tree removal rates in urban settings were the subject of a thorough review by the US Forest Service (Hilbert et al., 2019). That study draws on dozens of urban tree removal studies, and identifies 33 papers which offer sufficient detail to specify an annual removal rate; the study provides summarised annual morality ranges, reproduced here in Table 2. The report identifies high and low loss rates by quartile, and distinguishes between studies which track a planting cohort (i.e. a set of trees that is planted and monitored) and studies which consider an extant population of mixed ages, referred to as 'repeat inventory' studies by Hilbert et al. (2019).

The range of removal rates are not explicitly linked to specific management approaches, and therefore only represent outcomes of the various arrangements in place in the studied locations examined by Hilbert et al., (2019). It also represents a set of mortality studies that are configured differently to ours, in that we have a finite, modelled population with inbuilt replacement and an even distribution of age cohorts.

Our study therefore adopts a set of possible removal rate inputs that represent the approximate range of removal rates observed in first to third quartiles of the 'Planting cohort – Establishment' and 'Post-establishment' groups studied in Table 2, above, as a plausible lower and upper bound of loss rates across the scenarios monitored, avoiding exceptional values. These range from 1.5 % to 9.33 %. To enable a computationally manageable modelling process, we model this range in steps, testing removal rates of 1,3,5,7 and 9 % for each of the three age cohorts (young, maturing and mature). We model all possible combinations of each of these removal rates, resulting in a set of 125 scenarios (5^3).

Collectively this set of scenarios represent the plausible potential consequences of a range of common governance approaches; we recognise that governance interventions will vary in impact between contexts, so rather than attempting to model specific approaches, we model a full spectrum of removal rates to identify thresholds at which trees successfully reach maturity in a way that canopy targets are met, and ecosystem services grow over time. This therefore helps determine the minimum standards of tree retention that tree protection and establishment programs must meet, using the governance approaches that they deem likely to be locally effective. It is important to recognise that due to the use of quartile 1, median and quartile 3 values (as shown in Table 2, above) these are not optimal removal results, nor do the worst modelled results reflect the worst possible results in practice.

2.6. Modelling a diversity of settings

Recognising the diversity of cities' tree populations, we ran the model for a wide variety of potential parameter sets. Table 3 offers specific details of how we applied of these variations. We modelled a range of starting age and species distributions, as well as management variables such as (1) how frequently planting works are undertaken to replace removed trees, (2) how closely-spaced trees are in their planting, which we term 'planting density', (3) whether removals are compensated by one-for-one planting, or by planting multiple trees, (4) whether trees are replaced like-for-like or with different species, (5) the starting age of new trees that are planted, recognising that some cities plant very small saplings while others invest in advanced stock, (6) we apply various levels of 'growth slowing' to the model, to imitate a future setting in which growth conditions are more challenging (for example due to an increasingly hostile climate or built form). We also have a few model runs which treat the maximum ages specified in our allometrics as a point at which a tree stops growing, but does not necessarily immediately need to be removed. For clarity, we refer to these differently-parameterised model runs for each of the five city analogues as 'variant cases', reserving the term 'scenario' for the 125 removal rate scenarios that the model tests in every run, whether for a base or variant case.

2.7. Model operation

The model, built in Python, starts by generating a tree population with the species and the age distributions for 2025, to create a consistent 'base' from which each scenario can run. Then, it loops through 125 different levels of different tree removal (as explained above), applying the specified removal rates for each year between 2025 and 2100 to each age cohort. Trees within the cohort are randomly selected to be removed and replaced by saplings. This enables us to know the age that each tree

Table 2
Tree removal rates observed across 33 studies, as originally reported by Hilbert et al. (2019).

Study type	Number of studies	Higher/lower end of range (where ranges were reported)	Min.	1st quartile	Median	3rd quartile	Max.
Repeat inventory	18	lower	0	1.57	2.28	3.02	30
		higher	0	1.59	2.59	3.3	30
Planting cohort (all)	21	lower	0.6	2.81	4.4	7.08	68.47
		higher	0.6	3.76	6.48	9.33	68.47
Planting cohort (establishment)	16	lower	1.25	3.96	6.6	9.33	68.47
		higher	2.74	5.02	7	10.43	68.47
Planting cohort (post-establishment)	8	lower	0.6	1.5	2.76	3.81	4.6
		higher	0.6	1.53	3.76	4.73	11.22

Table 3
Details of the 27 variant cases that were modelled, encompassing a wide range of potential urban settings and management regimes. Each variant set is run for four location analogues.

Variant explored	Default	Specific setting of variant runs
Species age distribution	Even spilt across young/ maturing/mature.	7 scenarios total: All young trees, Heavy skew (75 % of all trees, 12.5 % in other categories) to young, maturing and mature, Lighter skew (50 % in skewed catergory, 25 % in others) for young, maturing and mature.
Species distribution skew	Even mix of all ~ 20 species for each regional set is applied.	3 scenarios total: in these scenarios, three randomly-selected species form 60 % of the tree population, with 20 % share each. Species selected are different in each of the three runs.
Exclusion of exotic species	No species are excluded.	1 scenario in which 5 non-native species with invasive potential in Florida were excluded: *Cinnamomum camphora, Eriobotrya japonica, Koelreuteria elegans, Platycladus orientalis, Triadica sebifera*. (Note that this is not run for other regional analogues, only Orlando).
How many trees are planted to replace a removed tree	One replacement tree is planted.	3 scenarios total: two replacement trees for each tree, three replacement trees for each tree, or replacement trees of equivalent DBH to the tree removed. In all scenarios these apply to a maximum of 1500 tree plots total.
Frequency of replanting removed trees	There is an annual tree planting program, so any removed tree is replanted the next year.	3 scenarios total, in which trees are replanted every two, three or five years.
Age of trees planted	Trees are three years old at planting.	2 scenarios total, one in which planted trees are five years old, one where trees are seven years old.
Density of planting	10 trees/100 m of road (both sides).	2 scenarios total. One in which planting density is 15 trees/100 m, and one in which density is 20 trees/ 100 m.
Species used for replacement	The same species is replanted.	1 scenario in which a random, different species is replanted when a tree is removed.
Treatment of trees at maximum allometric age	Trees at the age limit of our allometric equations are assumed to be at senescence and are removed that year.	3 scenarios in which trees reaching their maximum age are not assumed to hit senescence for another 5, 10 or 20 years, at which point they are removed.
Growing conditions for trees	Trees grow as per observed allometric rates	3 scenarios in which tree DBH increases slower than allometric rates. Growth is 10,20 and 50 % slower in these scenarios, respectively.

would be at each year of the study. Then application of the allometric equations discussed above enables calculation of DBH (trunk thickness) based on age, which in turn enables us to calculate annual canopy per tree per year. The model produces intermediate 'debug plots' which, while primarily produced for validation purposes for our team rather than as results, also offer a helpful visualisation of each individual tree's growth, replacement and regrowth over the study period. Supplementary 1 shows an example of the growth plots for a subset of trees in a model run for New York, which can be examined to aid in understanding the 'behind the scenes' logic of the model.

To aid interpretation of results, canopy is also output as a %p value, representing the total streetscape canopy cover relative to the street area in the study (i.e. canopy area vs 18 ha). This is done separately for each of the four climate regions. Fig. 2 summarises this process.

A section of our model automates the interpretation and visualisation of results, first by summarising the 1000-row result of each scenario (e.g. summing total canopy), and then by identifying the relative impact of each scenario at 2055, and its eventual long-term outcome at the end of the century.

3. Results

Our results are organised in subsections; each subsection corresponds to one of our three research questions. As the model's outputs are considerable, we initially focus on one set of variant runs for simplicity, drawing in examples from the larger result set later in this section.

3.1. Impacts of removal rates on target attainment

Fig. 3 shows how different removal rates produce very different canopy outcomes in thirty years for one of the fairly typical variant runs, which skewed the population slightly towards maturing trees. In most settings, only a few removal rate scenarios successfully delivered 'fast growth' (>5%p per decade). By contrast, scenarios delivering near-zero growth or net losses were quite common. The difference between allometric equation sets is striking, with New York and Orlando delivering quite strong differences in both the amount of 'fast growth' results, and the total canopy cover delivered.

In the longer-term model runs, we see strong early growth in low-removal scenarios, slowing or even trending slightly downward toward the end of the century, likely reflecting the effects of senescence and the corresponding loss of very large canopy trees. Nevertheless, scenarios that hit or exceeded 'fast growth' in 2055 still produce the highest long-term canopy percentages, at approximately double the starting canopy of 2025. A plot for 2100 growth is included at Supplementary 2 for those interested in direct comparisons to Fig. 3.

In Fig. 4, we examine both the 30-year and 75-year impacts of removal rates for the same run. The highest-performing scenarios all have a mature-tree mortality rate of 1 %; the cluster of positive results towards the centre of the plot mostly have a mature-tree mortality of 3 %; and negative results show a consistent mature-tree mortality of 5–9 %. In the top-right quadrant, slightly elevated removal rates of young and maturing trees was associated with better performance in the years between 2055 and 2100 in some scenarios, reflecting a longer process of maturation from a lower average age (which we discuss further below) which enabled sustained growth as fewer trees hit senescence (albeit still never achieving the results of the best scenarios, which retreat slightly from their peak as trees hit senescence but still exceed the total performance of the middle-cluster results).

By way of context, Table 4 offers an example of how this model's output might be used to understand the limits for tree removal required for achievement of a given canopy target, after a city applies the model to its tree population, using local allometrics exist for its urban trees. As this study focuses on street tree canopy, we demonstrate how different amounts of contribution of non-street canopy (on private property and in parks) will vary the targets that must be pursued using street canopy



Fig. 2. Summary of the model's operation. Code for the model is freely available in our Code Availability Statement.

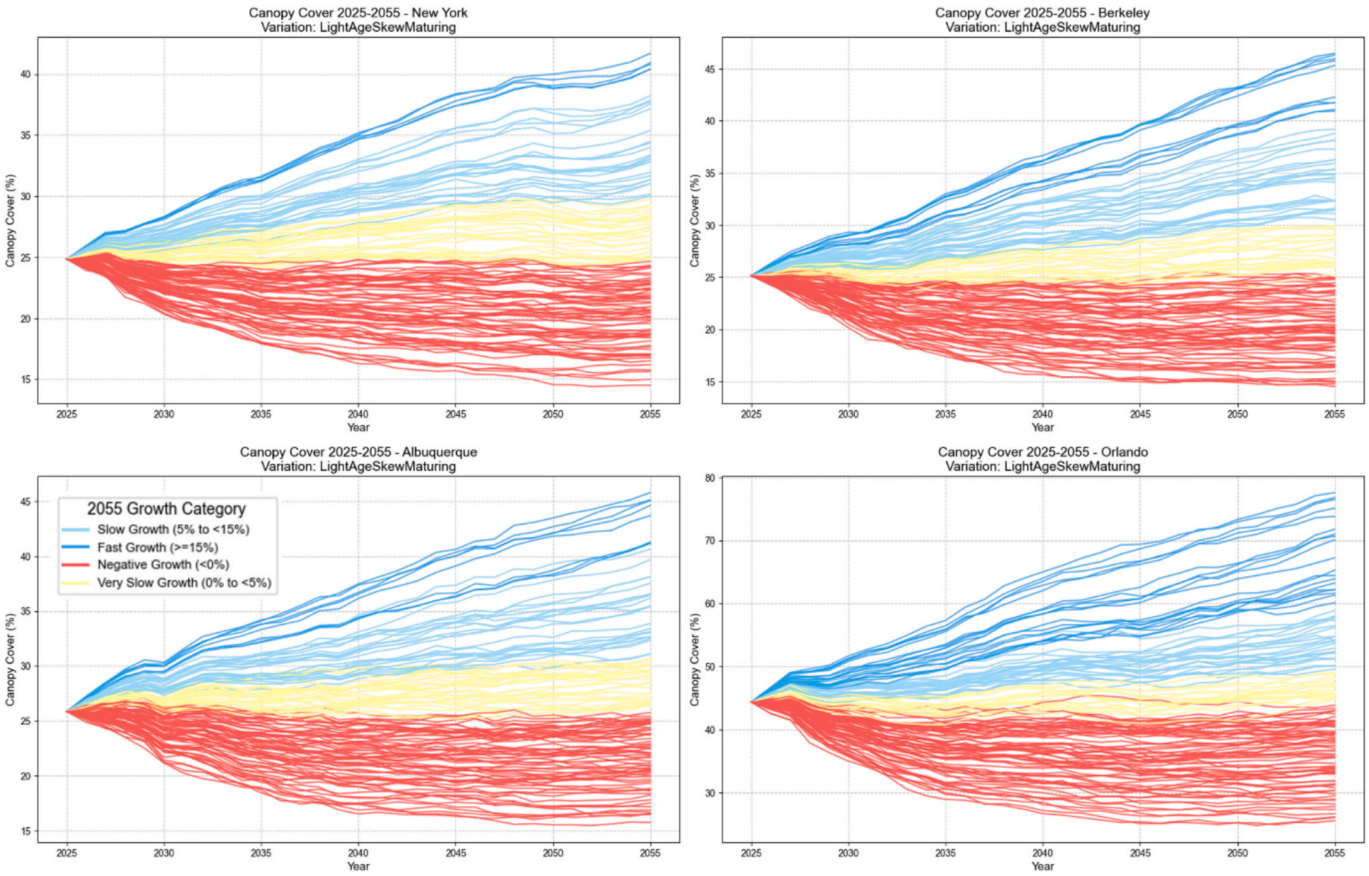


Fig. 3. Canopy trends observed when base case assumptions are modelled across 125 removal rate scenarios, across four hypothetical locations using the tree species composition and allometrics typical to the listed cities.

and the implications this has for removal tolerances.

3.2. Importance of each age cohort to overall canopy change

Fig. 5 offers deeper insight into the relative impact that removal rates by cohort had on canopy cover. The very strong negative correlation observed between mature tree removal rates and canopy increase ($r < -0.83$ in all locations) makes it clear that retaining large trees is vital to delivering the higher growth rates we observed in this study. Maturing tree retention was also correlated, albeit less strongly, while young tree removal rates were less determinative of achieving canopy increases.

3.3. Canopy outcomes under varied starting conditions

In many cases, variations to the model's parameters produced only moderate, quite predictable changes, but with a similar pattern, as is visible in the top panels of Fig. 6. Scenarios with reduced growth rates or intermittent planting (as in the top panels) produced weaker results than scenarios without these penalties, with lower overall canopy achievement and fewer 'fast growth' scenarios, as expected.

In other instances, strong age skews toward young or mature trees produce very different results. However, even in these cases, the fundamental relationship between the best and the worst results was very strongly predicted by removal rates of mature and maturing street trees, with the best results strongly predicted by mature and maturing tree removals equal to or under 3 %, in ways which would likely be highly relevant to achievement of policy targets. For example, in the 'heavy skew' scenario, the initial impression is of universal, disastrous dropoff of canopy – but the best scenario runs successfully retain just under 70 % street canopy, enough to meet some overall canopy targets singlehandedly, whereas the worst results fall below 30 % street canopy. The converse is true for the run with a large proportion of very young trees, where these trees reach excellent levels of canopy in the best scenarios, but cap out around 20–50 % in all but the lowest tree removal rates.

The diversity of Fig. 6 above is brought into striking consistency when correlations are tested; the result is almost indistinguishable from the plots in Fig. 5, with mature trees exerting a powerful influence on

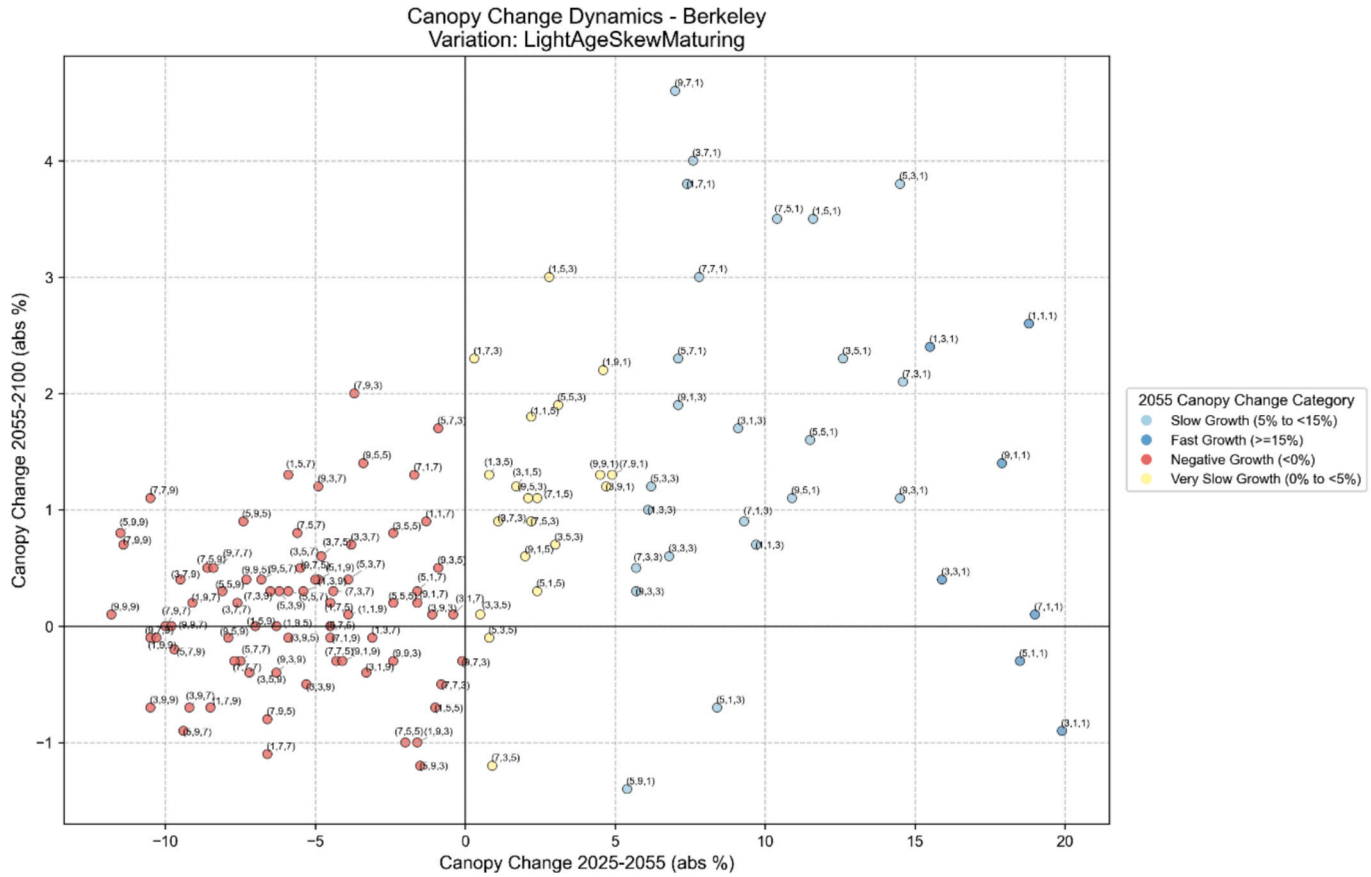


Fig. 4. Scatter plot showing individual scenario outcomes both for 2055 and 2100. This figure demonstrates the impact of different removal scenarios on medium term (30 yr) and long term (75 yr) canopy change. Bracketed numbers correspond to young, maturing and mature tree removal rates respectively. Colouration of dots pertains to 30-year impact (not 2100 result). Interestingly, in some scenarios which achieved rapid medium-term growth (e.g. 7,1,1) had slightly lower growth between 2055 and 2100 than scenarios with higher removal rates (e.g. 5,3,1), suggesting a 'ceiling' of canopy which these scenarios reach earlier. However, as is visible in Supplementary 2, scenarios with lower removal rates still showed the best canopy increase both overall and in the medium term.

final outcomes in all scenarios (p-values ranged from −0.71 to −0.8). A plot showing these correlations is available at Supplementary 3. The consistency of this result across such diverse input parameters indicates a consistently powerful influence of tree removal rates over canopy outcomes.

In a few variations that we modelled, these relationships were less robust, though still noticeable. The group of model runs in which removal of single trees triggered planting of multiple trees was quite resilient to higher removal rates, with a much higher proportion of positive results visible in Fig. 7. Surprisingly the 'DBH for DBH' run (which mandates planting of numerous small trees for a single mature tree removal) performed quite similarly to the 'one for two' run, which likely reflects that the population reaches its cap of 1500 trees fairly quickly in both scenarios, and then this population of new young trees drives strong growth in subsequent years. Nevertheless, the correlative relationship between removal rates and canopy outcomes remains strong, with an increased importance for maturing tree protection.

€One other area where results were notably different to the set was in the scenarios where trees were planted more densely. In these instances – where trees were planted at 1.5x and 2x the densities of the urban streets observed by Smart et al. (2020) – we see very high starting canopy and overall high results in Fig. 8, although the spread between these results is typical of the larger set – just within a higher range – as is notable in the correlative graphs. While removal rates still matter a great deal in these scenarios, it is clear that achieving a notional target of 60–80 % street canopy is much more possible using higher planting densities when removal rates are elevated.

Supplementary 4 includes a full, tabulated summary all variation results for all scenarios in all cities. This result affirms our key finding: for dozens of model runs with widely varying parameters, the impact of mature tree removal rates is strikingly consistent.

4. Discussion

Our results show clearly that tree removal rates have a very strong, quantifiable influence on canopy outcomes in a very wide variety of contexts, and that limiting removal of maturing and mature trees is vital to delivering canopy increases in contexts analogous to those we modelled. Canopy outcomes are highly sensitive to increases in removal rate of these trees, with most scenarios showing that 'fast growth' (as defined by this study as 5 %p canopy per decade) only possible under a strict adherence to removal rates of 1 % or less for mature trees. Increasing mature tree removals to 3 % was seen to sharply reduce canopy cover achievements, both over a typical three-decade policy timeframe and in the longer term, in a wide range of potential settings. These impacts can be somewhat mitigated by high tree planting densities and/or rules that require multiple trees in compensation for single removals, but even in these cases results are considerably better when mature tree removal is at 3 % or less. While an exploration of specific policy mechanisms for tree retention is not in the scope of this study, it is clear that tools targeted towards protecting mature and maturing trees (and ensuring they receive adequate maintenance to prevent issues) are of utmost importance to successful canopy target attainment.

Given that median rates of tree removal in the studies reviewed by Hilbert et al. (2019) ranged between 2.28 % to 7 %, it's likely that many urban forests around the world will have low or negative progress on the

Table 4
This table was assembled using the Berkeley analogue base case run, and assumes 40% of all land is streets, and the location has a canopy target of 30% overall canopy cover (all land uses) by 2055. We show how locations with different local canopy cover in their non-street land uses (e.g. parks, private property) must deliver different amounts of street canopy in order to hit the target, and what that means for removal tolerances. *In this example a 52.5% contribution from street canopy was not possible and would require not only restrictions on removals but also other complementary approaches (e.g. improved planting density, enhanced species selection to optimise canopy, and/or programs to enhance private canopy).

Non-street canopy	Necessary street canopy cover to hit overall 30 % cover	Age-cohort removal settings which met this target	Removal tolerance for maturing street trees to 2055	Removal tolerance for mature street trees to 2055
Low: (15 %)	52.5 %	None	<1%*	<1%*
Medium: (25 %)	37.5 %	(1,1,1), (3,1,1), (7,1,1), (5,1,1), (9,1,1), (1,3,1), (3,3,1), (5,3,1), (7,3,1), (9,3,1), (3,5,1), (1,5,1), (5,5,1)	3–5 %	1 %
High (40 %)	15 %	All scenarios except (5,9,9), (7,9,9) and (9,9,9)	7 %	7 %

extent of canopy cover. The subtle yet powerful effect of tree loss has the potential to negate the efforts of proactive urban forest managers seeking to increase the ecosystem services provided by canopy cover. Given the myriad threats to street trees in urban settings, ranging from construction to pruning and from resident complaints to root compaction, this is a striking finding – urban streets pose an array of risks to trees, to the point that healthy tree removals are perhaps somewhat normalised, but there is not much leeway for losses, especially once trees pass their establishment period. Even a few percentage points of upward creep in mature tree removal rates can result in very poor canopy results, if sustained. This study builds on similar findings from previous studies that demonstrate the vital nature of mature trees to ecosystem service delivery, for example the CityTree model developed by Rötzer et al. (2019) which shows that the benefits of older trees tend to multiply over time, with well-established individual trees offering many times the services of a young tree.

This finding highlights that cities seeking to meet canopy targets must have close regard for the maximum removal tolerances of their tree populations, and strive to stay well within these limits. Quantifying these limits using locally-specific canopy modelling is ideal, although at least a heuristic interpretation of our findings may serve in settings where modelling is unfeasible due to a lack of local allometric information and/or resources. Of course, no city has absolute control of removals, and severe storms or disease outbreaks may temporarily elevate removal rates, but maintaining a low overall rate of mature tree loss is demonstrably vital to canopy target achievement in many urban settings.

Achieving this level of control is clearly beyond the sole power of small municipal teams that tend to have responsibility for delivering urban forest strategies, who may be highly committed to tree protection, but lack substantial influence over the processes of property development, risk management and utility provision that can result in demands for avoidable removals of healthy trees. These teams also may not have the funding and/or authority to modify heavily engineered public spaces in ways that create sufficient soil volumes and stormwater infiltration opportunities to optimise tree health and maximise resilience to storm damage, pests and diseases. We encourage a recognition of tree protection as a challenge of broader urban landscape governance, requiring a coordinated policy, funding and regulatory response that treats mature and maturing trees as important public infrastructure.

In practical terms, this means tree retention requires the support and interest of a range of influential built environment stakeholders (such as property developers, road engineers, architects and construction contractors) many of whom may not currently consider tree protection to be either (a) important or (b) their responsibility (Kirkpatrick et al., 2013) While some of these individuals and organisations may altruistically exercise their power to protect trees, others will have overriding concerns that conflict with this behaviour. Accordingly, creation of a governance environment that strongly discourages avoidable tree removals in order to keep losses within defined limits, and embeds appropriate tree replacement responses in cases of unavoidable removal, is a basic prerequisite to the successful maintenance of an urban landscape that has adequate canopy cover. In this challenge, our model is a mechanism by which the need for these governance reforms can be evidenced, because it shows how vulnerable canopy target achievements are to even slightly elevated rates of removal.

Given that older trees may face higher rates of removal requests due to risks of root intrusion and limb drop (including real risks, and incorrectly overestimated risks), our findings also suggest a need for increased investment in tree maintenance, and community engagement to avoid unnecessary premature removals and/or excessive pruning of large mature trees (Fröhlich et al., 2024; Masini et al., 2023).

In cities pursuing infill development to improve housing supply, this finding is especially important for two reasons. Firstly, infill development tends to produce losses of private-realm canopy, as apartments displace gardens and back yards. This increases the importance of public-realm canopy cover (e.g. streetscape canopy, as quantified in this study), thereby making it crucial to deliver canopy targets (Croeser et al., 2020b). In instances where streets are the only source of trees, very high rates of canopy cover are necessary if suburbs hope to meet typical municipal canopy targets (e.g. 30 %, as proposed by Konijnendijk(2022), or 40 % (Ziter, (2019)); we demonstrate some examples of these high potential canopy requirements in Table 4 above. Secondly, intensification of housing and commercial development also places pressure on the public realm's trees, just as they become especially precious given the removal of their backyard counterparts(Hilbert et al., 2019b; Morgenroth et al., 2017). This requires an especially strong focus on tree retention, both during the design and construction of new buildings to ensure that driveways, gantries and underground infrastructure trenches are not placed in ways that require removal or heavy limb/root pruning of street trees.

This study finds that young tree retention is less impactful on long-term canopy outcomes than we had expected, particularly in longer-term variations of the model run, and for those with large populations of young trees. It appears that young tree removal rates of up to 7–9 % can be tolerated if replanting is carried out consistently, and removal is sharply limited once trees pass five years from planting. Hilbert et al. (2019) found 16 studies examining tree establishment failure, and the median removal rate across these studies was approximately 7 %, reflecting the particular vulnerability of young trees in urban settings. It is heartening that this management challenge, while important, is not severe enough to obstruct canopy progress alone. However, it is notable that our model uses randomisation to select which young trees are removed each year; in real-life settings, some contexts (e.g very hot microclimates, high rates of vandalism) will mean that young trees are more likely to face repeat removals at a rate that the random model would not show. Addressing the underlying issues in these instances is an important prerequisite to ensuring that high removal rates do not prevent long-term progress.

While the finding that only a fraction of scenarios in most variants delivered 'fast' canopy increase may seem discouraging, the parameter inputs of this study were important, and there is much to feel hopeful about in these findings when keeping these key assumptions in mind.

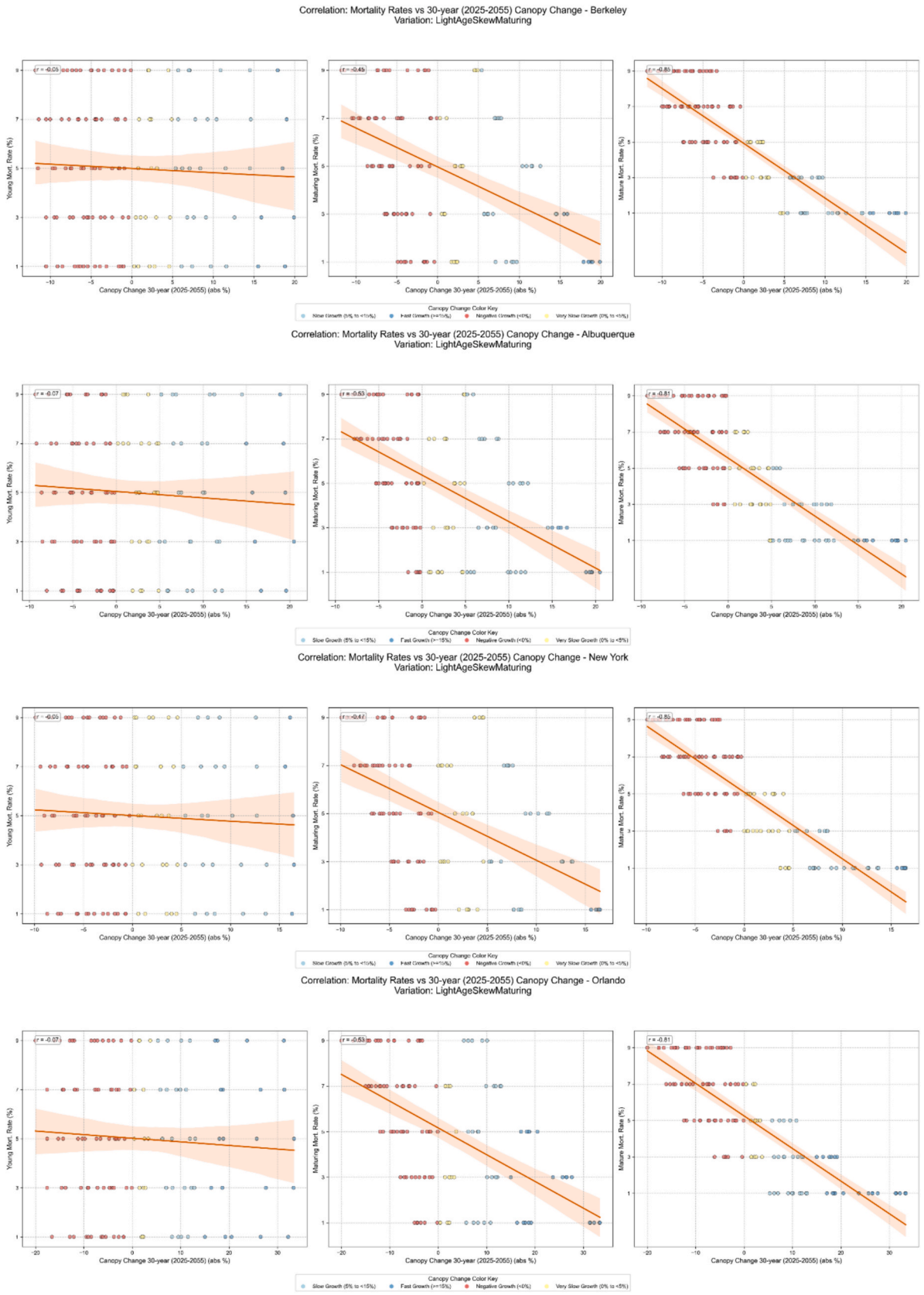


Fig. 5. Relationships between removal rate and canopy outcome, by age cohort, with Pearson Correlation Coefficient in the top left of each plot showing strength of correlations.

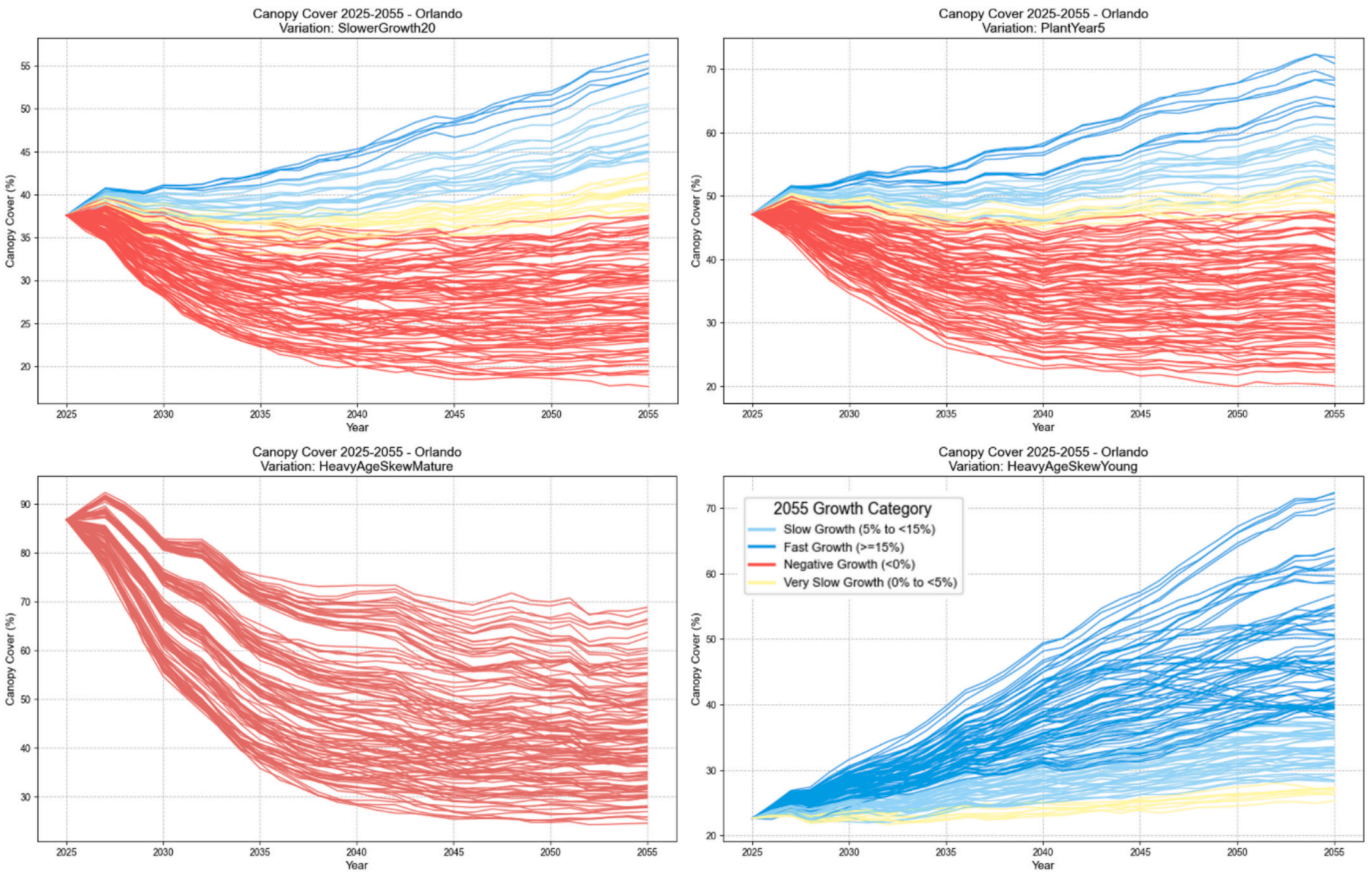


Fig. 6. Four variations of the 2055 outcome in Orlando, with diverse input assumptions. The top right panel involves a reduction of 20 % on tree growth rates, analogous to a more hostile climate or poorer planting conditions. The top right simulates a very intermittent planting schedule (every 5 years). The bottom panels show the impact of strong skews to the age distribution of the starting set of trees. The precise rules for each variation of the model are detailed in Table 3.

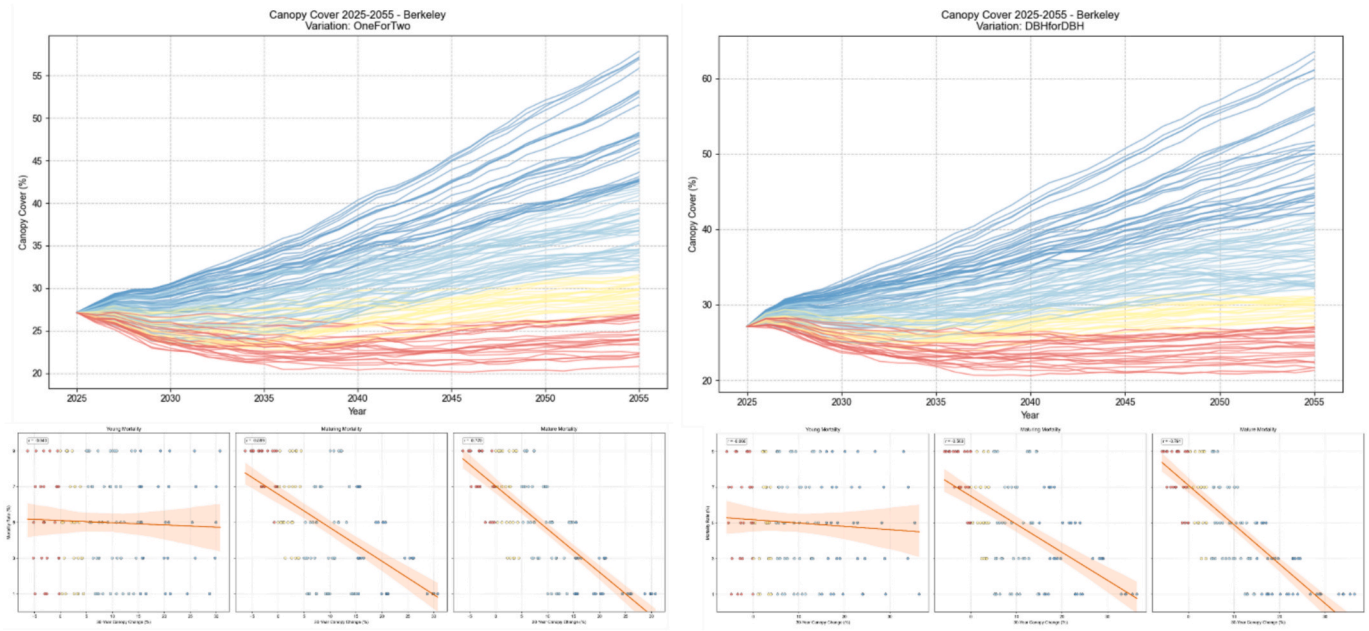


Fig. 7. Projected canopy growth to 2025, with correlation plots, for the 'DBH for DBH' and 'One for two' model runs, exploring impacts of different tree replacement rules.

Firstly, our study assumes a street is 'planted up' at a planting density of 10 trees per 100 m of street length, drawing on precedents documented in the literature; while the literature shows this to be typical, it is by no means dense (it equates to 20 m between tree plots). On a quiet residential street that is 18 m wide, there is space to plant on footpaths, between car parking spaces, and even to potentially create a small

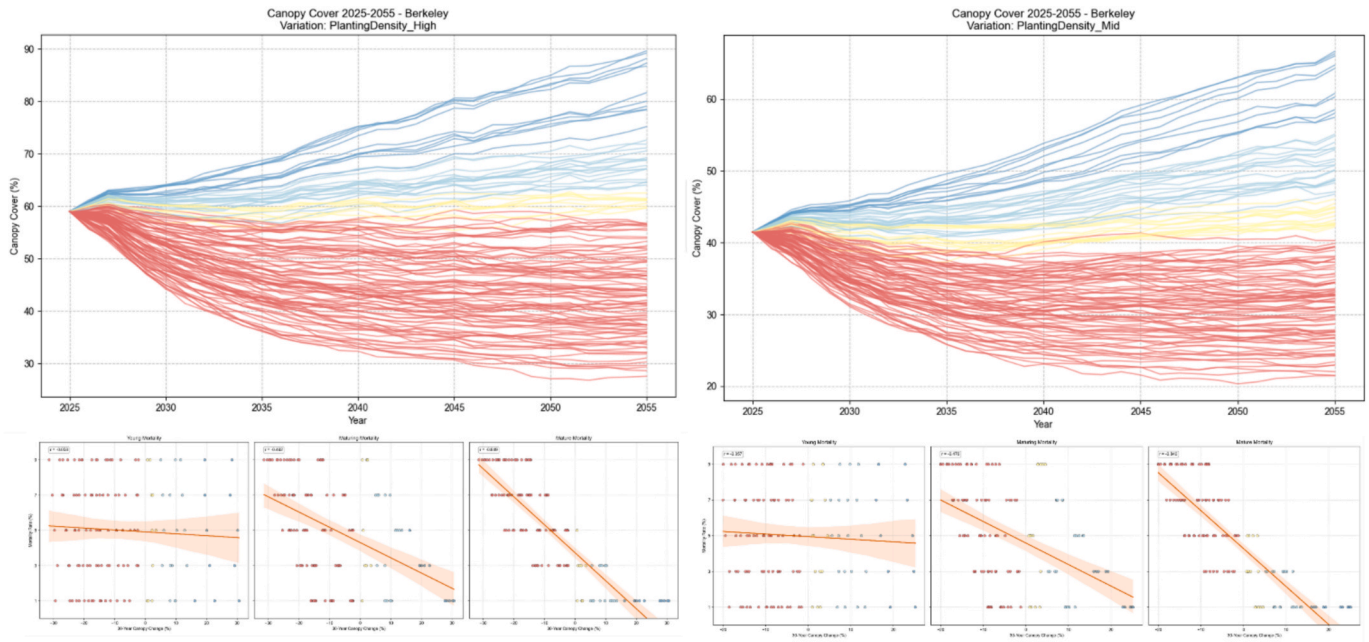


Fig. 8. Projected canopy growth to 2055, with correlation plots, for the two model runs which explored impacts of planting density. In the mid density example (right), the planting density is 1.5x normal, and in the 'high' version the density is doubled.

median strip enabling a third row of trees (especially in streets exceeding 20 m width, as is common in the US)(Millard-Ball, 2022). Even if trees are spaced 10 m apart and none are planted within 10 m of an intersection, it is possible that a triple row of trees would enable a planting density of 24 trees per 100 m of street; 2.4 times that of our assumed planting density, provided soil volume does not become a growth constraint. While the two variations of our model run which did assume higher planting density still offer a strong case for close management of tree removal, they also suggest that it is easier to meet challenging canopy targets with higher planting densities, especially in instances where mature and maturing tree removal rates cannot be reduced to rates approximating 1 %.

Secondly, our growth rates are based on observations of typical street trees. Typical street trees are not planted in very tree-friendly planting conditions; municipal budgets are often so limited that trees are simply placed in in a tree pit in a 1x1m hole in the footpath, with no enhancements of soil volume or quality, and no investments to facilitate stormwater infiltration. This means that it is quite likely that 'typical' tree growth rates, particularly in lower-rainfall areas, represent the growth rates of stressed trees; local planting conditions can produce considerable variance in growth rates (Czaja et al., 2020; Moser et al., 2017; Rötzer et al., 2019a; Vogt et al., 2015). Studies have shown substantially faster growth rates – up to double – for trees that have received these kinds of investments in their success (Grey et al., 2018). From this perspective, it is clear that cities must not only manage their tree removal rates carefully, but also engage in streetscape retrofits that enhance the planting density and growing conditions for trees, and select species sets that perform strongly in canopy delivery.

5. Limitations

The age limits discussed in the allometric formulae in this study required that we assumed tree removal for any tree that met the equation's maximum age limit. While this does indicate that the large US Forestry Service study did not find enough trees above this age to include them in their equation, and therefore this kind of tree age is rare, it also may serve to underestimate the absolute potential tree age, DBH and canopy achievable possible in well-managed urban forests where very mature trees of long-lived species may be kept alive in a way that they reach exceptional 'outlier' sizes. Given that centuries-old trees are not a common occurrence in urban forests, we are not concerned that this underestimation limits the general applicability of our results. Nevertheless, we must acknowledge that that a more precise model would distinguish actively between removals due to senescence, and removals due to other management decisions, ideally based on a more thoroughly-evidenced 'likely maximum average street tree age' rather than bundling senescence and management removals into to the 'mature' removal rate together.

This study does not explicitly account for a number of less predictable factors that can serve to increase tree removal rates. We could not account for the specific effects of a wide range of unpredictable destructive events such as violent storms, pest insect or disease outbreaks, vandalism and extended droughts – each of which can accelerate removal. Tree limb pruning, particularly heavy pruning mandated by highly conservative risk management frameworks to maintain cable clearances or avoid other liabilities, was also not accounted for; this can be a significant source of lost canopy, and potentially also shorten tree lifespans (Suchocka et al., 2021).

By modelling dozens of variations on key urban forest parameters, and identifying a consistently strong negative correlation between mature/maturing tree removal and strong canopy results, we aim to accommodate the variety of urban settings in which canopy must be maintained as an integral part of a healthy urban landscape. However, none of these model runs will perfectly approximate the specific setting of any given city, and that is not their function. Our primary intention is to demonstrate how a precise local street tree removal tolerance *can* be defined, and examine general results across a suite of model runs for a generic suburb in our four city analogues. We do not propose to perfectly define what tree removal tolerances might apply to a given location, and would encourage any user of this method to modify the model's inputs to reflect the age, size and species distribution of their street tree population, as well as local tree replacement rules and allometrics.

5.1. Frontiers for development

The model developed for the purposes of this study is tightly scoped, and lacks the user interfaces and varied features of practitioner tools like iTree, nor does it model tree physiological processes to produce the more

sophisticated, locally-tailored ecosystem service estimates that CityTree can be used to produce (Reitberger et al., 2025; Rötzer et al., 2019a). In this study the model used city-scale allometrics and is strongly shaped by the assumptions around species and age distribution discussed above. While existing tools do not have the functionality of modelling a planted-out urban forest (with the inbuilt assumption of tree replacement), this function could be integrated into these tools, and potentially refined for practitioner use. To enable cities to set locally-specific tree removal rate limits, a future iteration of this model could be built into existing systems to accommodate a number of input variables. Chief among these are (1) species-specific thresholds of maturity and decline, (2) growth rate modifiers for planting conditions and future climates, (3) the potential to input real tree inventories with known age distributions, (4) modifiable sliders showing impacts of management changes such as investments in planting condition improvements, greater planting densities, or changes in species distribution to optimise desired ecosystem service provision, (5) a degree of non-replacement of removed trees when planting sites prove unsuitable and (6) the baseline level of canopy in parks and private properties, enabling street canopy targets to be set in a way that delivers overall landscape canopy cover targets while accounting for context. By integrating these functions with the removal target-setting core of the model used in this paper, and then building it into a system like iTree or CityTree, quite sophisticated projections of tree removal management consequences could be developed.

6. Conclusion

Maintaining a stable population of large, healthy street trees is an important element of urban landscape management, but to date targets for limiting tree removal have not been established to support achievement of tree canopy targets. This study employed allometric modelling to show that, in a diversity of contexts, a general tolerance of around 1 % loss of mature trees in an urban forest annually is a good limit to pursue, ideally with no more than 3 % loss of maturing trees. Above this limit, our model indicates that common canopy increase targets become very difficult to meet. However, these rules of thumb should be approached tentatively; we model only US trees, and assume a set of planting densities and age/species distributions that is plausible but may differ from the local contexts of other cities. The more important contribution of this research is in demonstrating that firstly, modelling can be used to set a tree removal target that reflects local conditions, and secondly, it is likely that these targets need to be quite low in any city that hopes to achieve real progress on canopy cover. Given that cities often have removal rates well above our identified thresholds of failure, we argue that there is grounds for better-integrated urban governance for tree protection, engaging a greater diversity of the built environment stakeholders that make decisions that can result in avoidable removals of street trees.

CRediT authorship contribution statement

Thami Croeser: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Wolfgang W. Weisser:** Writing – review & editing, Methodology, Conceptualization. **Joe Hurley:** Writing – review & editing, Methodology, Conceptualization. **Thomas Rötzer:** Writing – review & editing, Validation, Methodology. **Leila Parhizgar:** Writing – review & editing, Writing – original draft, Validation, Methodology. **Qian (Chayn) Sun:** Writing – review & editing, Validation, Methodology. **Sarah A. Bekessy:** Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.landurbplan.2025.105484.

Data availability

All data produced in this study is freely available via Figshare – DOI https://doi.org/10.25439/rmt.29966033.

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